

Aditya Kaleeswar

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EDUCATION

Vellore Institute of Technology

Sep 2022 – Jul 2026

B.Tech in Computer Science · CGPA 8.41 — Katpadi, Vellore

Maharishi Vidya Mandir

Apr 2018 – Jul 2022

Senior Secondary (CBSE) · 96% (XII), 94% (X) — Chetpet, Chennai

EXPERIENCE

Research Intern

Dec 2025 – Present

UTRGV (AIM Lab) × VIT Vellore · Remote / Vellore

- **Engineered Cementitious Composites (ECC)**: built a recommender predicting the tensile strength & ductility of “bendable concrete” directly from mix-design parameters, to optimise mixes for cost vs. mechanical performance.
- Used **CatBoost** with **Mondrian Conformal Quantile Regression** for distribution-free, calibrated 80% prediction intervals — targeting $\approx 2.75\%$ strain capacity ($\approx 250\times$ conventional concrete) for seismic resilience.
- **Physics-informed crack assessment**: a computer-vision front-end (**YOLO-seg** + **SAM 2** segmentation, **Depth Anything V2** depth) with branch-splitting + fragment-linking recovers each crack’s geometry — arc length, angle, width, interactions — and mm-per-pixel calibration.
- That geometry feeds a **physics-informed neural network** used as an offline **solver** (one solution per a/W ratio via LEFM similarity), returning microsecond per-crack Mode-I stress-intensity (K_I) verdicts against fracture toughness.

[ECC preprint](#) · [ECC repo](#) · [PINN preprint](#) · [PINN repo](#)

Research Intern

Oct 2024 – Jun 2025

Samsung Research, Bangalore · Remote

- Built **BQ2Viz**, a multi-agent GenAI framework (LangGraph · LangChain · ChromaDB) that turns natural language into data visualizations via a profiling agent + a self-correcting code generator.
- Designed a **reasoning-based planner** for multi-file retrieval, inferring multiple distinct charts from a single abstract query.
- Benchmarked vs **NVAGENT** on **VisEval**: **84.57%** accuracy on complex multi-table queries; cut error rate to **11.5%** with a feedback-driven repair loop.

[NL→Data-Viz repo](#)

Software Engineer — Autonomous Domain (Senior Autonomous Member)

Apr 2024 – Present

Team Vyadh (Mars Rover Team), VIT · Vellore

- Built the autonomous navigation stack (**ROS 2 / Nav2 + SLAM**) with a **Frontier Exploration** algorithm to map unknown environments within dynamic GPS geofences.
- Real-time **obstacle & pit avoidance** using **Intel RealSense D455 + PCL**, with odometry path recovery and a PID alignment to compensate for wheel drift.
- Fused IMU + odometry with an **Extended Kalman Filter** to cut state-estimation noise; built a telemetry GUI for Astro-Bio streams (CO₂, IR, moisture). Team placed **13th at IRC** and **5th at IRDC**.

[Obstacle-avoidance repo](#) · [Rover-stack repo](#)

PROJECTS

HazMap — Autonomous Multi-Hazard Mapping Rover	Dec 2025 – May 2026
<i>ROS 2 · NVIDIA Jetson Orin Nano · EKF · GADEN · Bayesian Estimation · Python</i>	
▶ Hybrid edge–cloud autonomous rover that explores unknown environments and builds unified chemo-visual hazard maps, fusing LiDAR and gas-sensor data through an Extended Kalman Filter. ▶ Coverage planning: a modified C* algorithm incrementally builds a Reachability Connectivity Graph (RCG) and drives boustrophedon laps until the full navigable area is covered. ▶ Gas-source localization (ROS 2, validated in GADEN): CUSUM-based hotspot detection, real-time concentration mapping, and Bayesian probabilistic source estimation to pinpoint leak origins.	
preprint · hazpatrol repo · CGLS_gaden repo	
Alpha Stack — Natural Language → Production Codebases	Aug 2025 – Nov 2025
<i>Python · Multi-Agent Orchestration · RAG · NetworkX (DAGs) · Docker (sandbox) · CLI</i>	
▶ Hierarchical multi-agent system (Dominant–Worker model) that turns a natural-language prompt into a complete, production-ready, multi-file codebase. ▶ Tree-based DFS traversal propagates semantic context across files; an incremental Dependency Analysis Engine on Directed Acyclic Graphs (NetworkX) maps inter-file relationships in $O(k)$ time, preventing circular dependencies. ▶ A self-healing test → fail → diagnose → repair loop drives validation: 79.10% executability on the ProjectDev benchmark vs ChatDev’s 32.79%.	
preprint · repo	

PATENT

Force Sensor-Based Train Presence & State Detection	Indian Patent Application No. 202641007500
<i>ESP32 · Force-Sensitive Resistors (FSR) · Embedded C · Finite-State Machine · IoT</i>	
▶ Platform-level IoT system that passively detects and classifies a train’s state — approaching, stopped, pass-through, departing — and its direction of entry, with no train-side equipment. ▶ An array of Force-Sensitive Resistors under the track measures the train’s normal force; distributed ESP32 nodes do edge thresholding and send digital triggers to a central ESP32 running a finite-state machine that classifies state from trigger sequence and timing. ▶ Minimally invasive and low-cost (no track-circuit modifications), enabling platform automation such as automatic barriers, alerts, and announcements.	

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript (ES6+), SQL (Postgres), HTML/CSS
GenAI & ML: PyTorch, LangGraph, LangChain, RAG, Pinecone, ChromaDB, Ollama, OpenRouter, CatBoost, Conformal Prediction
Robotics & Vision: ROS 2 (Nav2), SLAM, OpenCV, PCL, YOLOv8, TensorRT, NVIDIA Jetson, ESP32
Full Stack & DB: React, Node.js, Express, MongoDB, MERN, REST APIs, Vite
Data & Algorithms: pandas, NumPy, NetworkX, SciPy, Matplotlib, scikit-learn, DuckDB
DevOps & Cloud: GCP, Docker, Git, Linux